Honeybees: Life Inside the Colony

Honeybees are social insects that live in colonies of tens of thousands of individuals. A single colony contains one queen, several thousand male drones during the warm months, and a large population of female worker bees. Although the western honeybee is the most widely kept species, several other honeybee species live across Asia, where the group is thought to have originated.

The queen is the only fully fertile female in the colony. After a mating flight early in her life, she stores sperm from many drones and can lay up to two thousand eggs a day during the peak of the season. Fertilised eggs develop into workers or new queens, while unfertilised eggs develop into drones.

Worker bees change jobs as they age. Young workers clean cells and feed larvae, slightly older bees build wax comb and process nectar, and the oldest workers become foragers that collect nectar, pollen, water and plant resin outside the hive. In summer a worker usually lives for about five to six weeks, but workers raised in autumn can survive through the winter.

Drones are male bees whose main role is to mate with queens from other colonies. They have no stinger and do not collect food. At the end of the season, when resources become scarce, workers often push the remaining drones out of the hive.

Foragers share the location of good food sources through the waggle dance. On the vertical surface of the comb, a dancer runs in a straight line while shaking her abdomen, then loops back and repeats. The angle of the straight run relative to vertical indicates the direction of the food relative to the sun, and the duration of the run indicates the distance.

Honey is made from nectar. Foragers carry nectar back in a special honey stomach, and house bees pass it between them, adding enzymes that break down sugars. The bees then fan their wings to evaporate water until the nectar is thick enough to store, and finally seal each cell with a cap of wax.

Beeswax is produced by glands on the underside of young workers' abdomens. Workers chew the small wax flakes and shape them into hexagonal cells. The hexagon is an efficient shape: it lets the bees store the most honey and brood while using the least amount of wax.

Honeybees keep the brood nest at a nearly constant temperature of about thirty-five degrees Celsius. In cold weather they cluster together and generate heat by shivering their flight muscles; in hot weather they spread water droplets on the comb and fan their wings to cool it by evaporation.

A colony reproduces by swarming. When a colony grows crowded, workers raise new queens, and the old queen leaves with roughly half of the workers. The swarm rests on a branch while scout bees search for a new home, and the scouts reach a decision by dancing for the sites they have found until one site attracts a clear majority.

Honeybees are important pollinators of many fruit, nut and seed crops. Colonies face several threats, including the parasitic varroa mite and the viruses it spreads, pesticide exposure, poor nutrition from a lack of diverse flowers, and the stress of being moved long distances for commercial pollination.